

18. Rectifiability and orthogonal projections

Besicovitch–Federer projection theorem

In this chapter we are going to give a characterization of rectifiable sets in terms of their projection properties. We have already seen in Theorem 16.2 that if E is an $\mathcal{H}^m$ measurable m -rectifiable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) > 0$, then $\mathcal{H}^m(P_V E) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$. The main result of this chapter is that for a purely m -unrectifiable $\mathcal{H}^m$ measurable set A with $\mathcal{H}^m(A) < \infty$ we have $\mathcal{H}^m(P_V A) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$. This deep result was proved first by Besicovitch [5] in the case $n = 2$, $m = 1$, and then by Federer [1] for general dimensions.

In this chapter m and n will be integers with $0 < m < n$.

18.1. Theorem. *Let A be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(A) < \infty$.*

(1) *A is m -rectifiable if and only if $\mathcal{H}^m(P_V B) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$ whenever B is an $\mathcal{H}^m$ measurable subset of A with $\mathcal{H}^m(B) > 0$.*

(2) *A is purely m -unrectifiable if and only if $\mathcal{H}^m(P_V A) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.*

Again these statements are equivalent and in view of Theorem 16.2 we only need to show that a purely m -unrectifiable $\mathcal{H}^m$ measurable set with finite $\mathcal{H}^m$ measure projects into zero $\mathcal{H}^m$ measure on almost all m -planes. This will be done through the lemmas 18.2–9. Roughly speaking the structure of the proof is the following: for a given $V \in G(n, n-m)$ we shall consider three subsets of A and show that they all have projection on $V^\perp$ of measure zero, and then we shall show that for almost every V one of the three alternatives defining these subsets occurs almost everywhere in A .

In Lemmas 18.2, 18.3, 18.4, 18.7 and 18.9 we shall always assume that A is an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(A) < \infty$. For the notation, recall 15.12. Our first lemma is an immediate consequence of Corollary 15.15.

18.2. Lemma. *Suppose A is purely m -unrectifiable. Let $\delta > 0$, $V \in G(n, n-m)$ and*

$$A_{1,\delta}(V) = \left\{ a \in A : \limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(a, r, V, s)) = 0 \right\}.$$

Then $\mathcal{H}^m(A_{1,\delta}(V)) = 0$.

18.3. Lemma. Let $\delta > 0$, $V \in G(n, n - m)$ and

$$A_{2,\delta}(V) = \left\{ a \in A : \limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(a, r, V, s)) = \infty \right\}.$$

Then $\mathcal{H}^m(Q_V(A_{2,\delta}(V))) = 0$.

Proof. Let $0 < M < \infty$. For all $a \in A_{2,\delta}(V)$ there are arbitrarily small $s > 0$ and some r , $0 < r < \delta$, such that

$$(1) \quad \mathcal{H}^m(A \cap X(a, r, V, s)) \geq M(rs)^m = M2^{-m} \mathcal{H}^m(Q_V X(a, r, V, s));$$

note that $Q_V X(a, r, V, s) = U(Q_V a, rs) \cap V^\perp$. We apply Vitali's covering theorem 2.2 in the m -plane $V^\perp$ to find $a_i \in A_{2,\delta}(V)$, $i = 1, 2, \dots$, and the corresponding numbers s_i and r_i satisfying (1) such that the m -balls $Q_V X(a_i, r_i, V, s_i)$ are disjoint and

$$\mathcal{H}^m\left(Q_V(A_{2,\delta}(V)) \setminus \bigcup_{i=1}^{\infty} Q_V X(a_i, r_i, V, s_i)\right) = 0.$$

Then

$$\begin{aligned} \mathcal{H}^m(Q_V(A_{2,\delta}(V))) &\leq \sum_{i=1}^{\infty} \mathcal{H}^m(Q_V X(a_i, r_i, V, s_i)) \\ &\leq \sum_{i=1}^{\infty} M^{-1} 2^m \mathcal{H}^m(A \cap X(a_i, r_i, V, s_i)) \leq M^{-1} 2^m \mathcal{H}^m(A). \end{aligned}$$

Letting $M \rightarrow \infty$ we have $\mathcal{H}^m(Q_V(A_{2,\delta}(V))) = 0$. $\square$

18.4. Lemma. Let $V \in G(n, n - m)$ and

$$A_3(V) = \{a \in A : \text{card}(A \cap (V + a)) = \infty\}.$$

Then $\mathcal{H}^m(Q_V(A_3(V))) = 0$.

Proof. Since $\text{card} = \mathcal{H}^0$, this is an immediate consequence of Theorem 7.7:

$$\begin{aligned} \int_{V^\perp}^* \text{card}(A \cap (V + y)) d\mathcal{H}^m y &= \int_{V^\perp}^* \mathcal{H}^0(A \cap Q_V^{-1}\{y\}) d\mathcal{H}^m y \\ &\leq c\mathcal{H}^m(A) < \infty, \end{aligned}$$

whence $\text{card}(A \cap (V + y)) < \infty$ for $\mathcal{H}^m$ almost all $y \in V^\perp$, which is the same as our assertion. $\square$

We shall use the following general density theorem of Mickle and Radó [1]. The interest in it lies in the fact that Ψ may be very non-additive. For Borel measures one could say much more by Theorem 2.12. We shall actually apply it with $\mathbf{R}^n$ replaced by S^{n-1} and $\mathcal{L}^n$ by $\mathcal{H}^{n-1} \llcorner S^{n-1}$, but the same proof works also in this case.

18.5. Theorem. Let Ψ be a measure on $\mathbf{R}^n$ and E an $\mathcal{L}^n$ measurable set with $\Psi(E) = 0$. Then for $\mathcal{L}^n$ almost all $x \in E$,

$$\limsup_{r \downarrow 0} r^{-n} \Psi(B(x, r)) = 0 \quad \text{or} \quad = \infty.$$

We first prove a lemma.

18.6. Lemma. Let Ψ be a measure on $\mathbf{R}^n$, F a closed subset of $\mathbf{R}^n$ and δ and M positive numbers. If

$$\Psi(B(x, r)) \leq Mr^n \quad \text{whenever } 0 < r < \delta \text{ and } B(x, r) \cap F \neq \emptyset,$$

then

$$\lim_{r \downarrow 0} r^{-n} \Psi(B(x, r) \setminus F) = 0 \quad \text{for } \mathcal{L}^n \text{ almost all } x \in F.$$

Proof. Let $x \in F$ and $0 < r < \delta/5$. Put

$$s_y = d(y, F)/2 \quad \text{for } y \in B(x, r) \setminus F.$$

Then $0 < s_y \leq r/2$ and $B(y, s_y) \subset B(x, 2r) \setminus F$. By Theorem 2.1 there is a countable set $S \subset B(x, r) \setminus F$ such that the balls $B(y, s_y)$, $y \in S$, are disjoint and

$$B(x, r) \setminus F \subset \bigcup_{y \in S} B(y, 5s_y).$$

Hence

$$\Psi(B(x, r) \setminus F) \leq 5^n M \sum_{y \in S} s_y^n \leq 5^n M \alpha(n)^{-1} \mathcal{L}^n(B(x, 2r) \setminus F).$$

By the Lebesgue density theorem 2.14,

$$\lim_{r \downarrow 0} r^{-n} \mathcal{L}^n(B(x, 2r) \setminus F) = 0 \quad \text{for } \mathcal{L}^n \text{ almost all } x \in F,$$

which proves the lemma. □

To prove Theorem 18.5 we may assume E to be closed. For $j = 1, 2, \dots$, set

$$F_j = \{x \in E : \Psi(B(x, r)) \leq jr^n \quad \text{for } 0 < r < 1/j\}.$$

Then each F_j is closed and

$$\{x \in E : \limsup_{r \downarrow 0} r^{-n} \Psi(B(x, r)) < \infty\} = \bigcup_{j=1}^{\infty} F_j.$$

Hence it suffices to show that $\lim_{r \downarrow 0} r^{-n} \Psi(B(x, r)) = 0$ for $\mathcal{L}^n$ almost all $x \in F_j$. For this we use Lemma 18.6. If $B(x, r) \cap F_j \neq \emptyset$, there is $y \in F_j$ such that $B(x, r) \subset B(y, 2r)$ and so $\Psi(B(x, r)) \leq 2^n j r^n$ provided $r < 1/(2j)$. Hence, as $\Psi(E) = 0$, Lemma 18.6 yields

$$\lim_{r \downarrow 0} r^{-n} \Psi(B(x, r)) = \lim_{r \downarrow 0} r^{-n} \Psi(B(x, r) \setminus F_j) = 0$$

for $\mathcal{L}^n$ almost all $x \in F_j$. □

Now we come to the most difficult lemma.

18.7. Lemma. *Let $\delta > 0$. The following holds for $\gamma_{n, n-m}$ almost all $V \in G(n, n-m)$. For $\mathcal{H}^m$ almost all $a \in A$ either*

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(a, r, V, s)) = 0$$

or

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(a, r, V, s)) = \infty$$

or

$$(A \setminus \{a\}) \cap (V + a) \cap B(a, \delta) \neq \emptyset.$$

Proof. We first prove the lemma in the case $m = n - 1$ and then essentially reduce the general case to this by some integral geometry. There are some measurability problems which we leave to the reader. In fact, we may and shall assume A to be σ -compact in view of Theorem 1.10 (1), and in this case the required measurabilities are easier to check.

We shall prove, in the case $m = n - 1$, that given $\delta > 0$ and $a \in \mathbf{R}^n$ we have for $\gamma_{n,1}$ almost all $L \in G(n, 1)$ either

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{1-n} \mathcal{H}^{n-1}(A \cap X(a, r, L, s)) = 0$$

or

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{1-n} \mathcal{H}^{n-1}(A \cap X(a, r, L, s)) = \infty$$

or

$$(A \setminus \{a\}) \cap (L + a) \cap B(a, \delta) \neq \emptyset.$$

The statement of Lemma 18.7 (for $m = n - 1$) follows from this by Fubini's theorem.

To simplify notation assume $a = 0$. For $\theta \in S^{n-1}$ and $B \subset S^{n-1}$ let

$$L_\theta = \{t\theta : t \in \mathbf{R}\} \quad \text{and} \quad L(B) = \bigcup_{\theta \in B} L_\theta.$$

Define a measure Ψ by

$$\Psi(B) = \sup_{0 < r < \delta} r^{1-n} \mathcal{H}^{n-1}(A \cap B(r) \cap L(B)) \quad \text{for } B \subset S^{n-1},$$

and set

$$C = \{\theta \in S^{n-1} : (A \setminus \{0\}) \cap B(\delta) \cap L_\theta \neq \emptyset\}.$$

Since A is σ -compact, so is C . Letting $E = S^{n-1} \setminus C$ we have $\Psi(E) = 0$ by the definitions of Ψ and C . Hence by the obvious analogue of Theorem 18.5 for $\mathcal{H}^{n-1}$ on S^{n-1} we obtain for $\mathcal{H}^{n-1}$ almost all $\theta \in S^{n-1}$ either

$$\limsup_{t \downarrow 0} t^{1-n} \Psi(S^{n-1} \cap B(\theta, t)) = 0$$

or

$$\limsup_{t \downarrow 0} t^{1-n} \Psi(S^{n-1} \cap B(\theta, t)) = \infty$$

or

$$\theta \in C.$$

We have for any $x, \theta \in S^{n-1}$ with $x \cdot \theta \geq 0$

$$d(x, L_\theta) \leq |x - \theta| \leq 2d(x, L_\theta).$$

This gives

$$X(0, r, L_\theta, s) \subset B(r) \cap L(S^{n-1} \cap B(\theta, 2s)) \setminus \{0\} \subset X(0, r, L_\theta, 3s).$$

Taking into account the definition of Ψ we see that the three alternatives proved give the three alternatives desired. Thus we have proven Lemma 18.7 in the case $m = n - 1$.

To continue we assume $m < n - 1$ and we first observe that the above proof gives a little more. Namely, assuming A to be σ -compact with $\mathcal{H}^m(A) < \infty$ we have for $\gamma_{m+1,1}$ almost all $L \in G(m+1, 1)$ either

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap B(r) \cap (X^{m+1}(0, L, s) \times \mathbf{R}^{n-m-1})) = 0$$

or

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap B(r) \cap (X^{m+1}(0, L, s) \times \mathbf{R}^{n-m-1})) = \infty$$

or

$$(A \setminus \{0\}) \cap (L \times \mathbf{R}^{n-m-1}) \cap B(\delta) \neq \emptyset.$$

Here

$$X^{m+1}(0, L, s) = \{x \in \mathbf{R}^{m+1} : d(x, L) < s|x|\}.$$

To get this statement apply Theorem 18.5 on S^m defining

$$L(B) = \bigcup_{\theta \in B} (L_\theta \times \mathbf{R}^{n-m-1}) \subset \mathbf{R}^n, \quad B \subset S^m,$$

$$C = \{\theta \in S^m : (A \setminus \{0\}) \cap (L_\theta \times \mathbf{R}^{n-m-1}) \cap B(\delta) \neq \emptyset\},$$

$$\Psi(B) = \sup_{0 < r < \delta} r^{-m} \mathcal{H}^m(A \cap B(r) \cap L(B)) \quad \text{for } B \subset S^m.$$

Otherwise the proof is the same.

So far we have been looking only at $(n-m)$ -planes of the form $L \times \mathbf{R}^{n-m-1}$, $L \in G(m+1, 1)$, but replacing $\{0\} \times \mathbf{R}^{n-m-1}$ by an arbitrary $W \in G(n, n-m-1)$ we have the same for the planes $L+W$ where L is a line in $W^\perp$. Every $V \in G(n, n-m)$ can be represented in this form, and if $L+W$ has some property for every W and almost all lines L in $W^\perp$, then, by simple analysis performed below, almost every $V \in G(n, n-m)$ has the same property. However, the sets $X^{m+1}(\cdot) \times \mathbf{R}^{n-m-1}$ are not exactly the cones of Lemma 18.7, but they are sufficiently closely related to them by the next lemma. For the details we need some notation.

Let

$$V = \{0\} \times \mathbf{R}^{n-m} \in G(n, n-m)$$

and for $0 < s < 1$, $j \in \{m+1, \dots, n\}$,

$$Z(j, s) = \left\{x \in \mathbf{R}^n : \sum_{i=1}^m x_i^2 < (s^2/(1-s^2)) x_j^2\right\}.$$

Note that

$$Z(m+1, s) = X^{m+1}(0, L_{m+1}, s) \times \mathbf{R}^{n-m-1}$$

where L_{m+1} is the x_{m+1} -axis in $\mathbf{R}^{m+1}$. We also define s^* , $0 < s^* < 1$, by

$$s^{*2}/(1-s^{*2}) = (n-m)s^2/(1-s^2).$$

In order to complete the proof of Lemma 18.7 we need two more lemmas.

18.8. Lemma. $\bigcup_{j=m+1}^n Z(j, s) \subset X(0, V, s) \subset \bigcup_{j=m+1}^n Z(j, s^*)$.

Proof. This follows immediately from the definition of $Z(j, s)$ and the fact

$$\begin{aligned} X(0, V, s) &= \left\{ x \in \mathbf{R}^n : \sum_{i=1}^m x_i^2 < s^2 \sum_{i=1}^n x_i^2 \right\} \\ &= \left\{ x \in \mathbf{R}^n : \sum_{i=1}^m x_i^2 < (s^2/(1-s^2)) \sum_{i=m+1}^n x_i^2 \right\}. \quad \square \end{aligned}$$

18.9. Lemma. Let $\delta > 0$ and $j \in \{m+1, \dots, n\}$. For θ_n almost all $g \in O(n)$ either

- (1) $\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap B(r) \cap gZ(j, s)) = 0$ or
- (2) $\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap B(r) \cap gZ(j, s)) = \infty$ or
- (3) $(A \setminus \{0\}) \cap g(V) \cap B(\delta) \neq \emptyset$.

Proof. We give the proof in the case $j = m+1$. Let χ be the characteristic function of those $g \in O(n)$ for which none of the properties (1)–(3) hold. Assuming again A to be σ -compact, χ is a Borel function. We identify, in the obvious way,

$$O(m+1) = \{g \in O(n) : g|_{\{0\}} \times \mathbf{R}^{n-m-1} \text{ is the identity}\}.$$

Since $gZ(m+1, s) = X^{m+1}(0, gL_{m+1}, s) \times \mathbf{R}^{n-m-1}$ for $g \in O(m+1)$, we obtain from what we have proved so far

$$(4) \quad \int_{O(m+1)} \chi d\theta_{m+1} = 0.$$

For any $h \in O(n)$, $h^{-1}(A)$ satisfies the same hypothesis as A . As the characteristic function corresponding to $h^{-1}(A)$ is $g \mapsto \chi(h \circ g)$, we obtain

$$\int_{O(m+1)} \chi(h \circ g) d\theta_{m+1} g = 0.$$

By the invariance of θ_{m+1} we have for any $g \in O(m+1)$

$$\int_{O(n)} \chi(h) d\theta_n h = \int_{O(n)} \chi(h \circ g) d\theta_n h.$$

Using these formulas and Fubini's theorem, we infer

$$\begin{aligned}\int_{O(n)} \chi(h) d\theta_n h &= \int_{O(m+1)} \int_{O(n)} \chi(h) d\theta_n h d\theta_{m+1} \\ &= \int_{O(m+1)} \int_{O(n)} \chi(h \circ g) d\theta_n h d\theta_{m+1} g \\ &= \int_{O(n)} \int_{O(m+1)} \chi(h \circ g) d\theta_{m+1} g d\theta_n h = 0,\end{aligned}$$

which proves Lemma 18.9. $\square$

Combining Lemmas 18.8 and 18.9 we get for θ_n almost all $g \in O(n)$ either

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(0, r, gV, s)) = 0$$

or

$$\limsup_{s \downarrow 0} \sup_{0 < r < \delta} (rs)^{-m} \mathcal{H}^m(A \cap X(0, r, gV, s)) = \infty$$

or

$$(A \setminus \{0\}) \cap g(V) \cap B(\delta) \neq \emptyset.$$

This holds for any $a \in \mathbf{R}^n$ in place of the origin. Hence Lemma 18.7 follows by the definition of $\gamma_{n,n-m}$. $\square$

We can now finish the proof of Theorem 18.1. Let $V \in G(n, n-m)$ and $\delta > 0$. We use the notations $A_{1,\delta}(V)$, $A_{2,\delta}(V)$ of Lemmas 18.2 and 18.3 and also let

$$A_{3,\delta}(V) = \{a \in A : (A \setminus \{a\}) \cap (V + a) \cap B(a, \delta) \neq \emptyset\}.$$

By Lemma 18.7 we have for $\gamma_{n,n-m}$ almost all $V \in G(n, n-m)$,

$$(1) \quad \mathcal{H}^m(A \setminus (A_{1,\delta}(V) \cup A_{2,\delta}(V) \cup A_{3,\delta}(V))) = 0.$$

We shall show that if $V \in G(n, n-m)$ is such that (1) holds for all $\delta \in S = \{1/i : i = 1, 2, \dots\}$, then $\mathcal{H}^m(Q_V A) = 0$. This will clearly prove Theorem 18.1.

First by Lemma 18.2, $\mathcal{H}^m(A_{1,\delta}(V)) = 0$, whence (1) implies

$$\mathcal{H}^m(A \setminus (A_{2,\delta}(V) \cup A_{3,\delta}(V))) = 0.$$

Secondly, $\bigcap_{\delta \in S} A_{3,\delta}(V) \subset A_3(V)$ with $A_3(V)$ as in Lemma 18.4 and

$$\begin{aligned} \bigcup_{\delta \in S} (A \setminus (A_{2,\delta}(V) \cup A_{3,\delta}(V))) &\supset A \setminus \left(\left(\bigcup_{\delta \in S} A_{2,\delta}(V) \right) \cup \left(\bigcap_{\delta \in S} A_{3,\delta}(V) \right) \right) \\ &\supset A \setminus \left(\bigcup_{\delta \in S} A_{2,\delta}(V) \cup A_3(V) \right), \end{aligned}$$

which gives

$$\begin{aligned} &\mathcal{H}^m \left(Q_V \left(A \setminus \left(\bigcup_{\delta \in S} A_{2,\delta}(V) \cup A_3(V) \right) \right) \right) \\ &\leq \sum_{\delta \in S} \mathcal{H}^m (A \setminus (A_{2,\delta}(V) \cup A_{3,\delta}(V))) = 0. \end{aligned}$$

But $\mathcal{H}^m(Q_V(\bigcup_{\delta \in S} A_{2,\delta}(V))) = 0$ by Lemma 18.3 and $\mathcal{H}^m(Q_V(A_3(V))) = 0$ by Lemma 18.4; thus $\mathcal{H}^m(Q_V A) = 0$ follows and Theorem 18.1 is proven. $\square$

Remarks on projections

18.10. (1) It is not known if the third alternative in Lemma 18.7 is really needed. This is essentially the problem mentioned in Remark 10.12. Another related old question is whether there exists a purely 1-unrectifiable compact subset of $\mathbf{R}^2$ with positive $\mathcal{H}^1$ measure intersecting every line in at most k points. The answer is not known for any integer $k \geq 2$.

(2) The preceding proof is essentially from Federer [3, § 3.3] but the presentation was greatly influenced by that of Ross [1]. Actually Federer proves a more precise result which is in a sense optimal. To explain this we first give a definition.

If μ is a measure on $\mathbf{R}^n$, a set $E \subset \mathbf{R}^n$ is called (μ, m) rectifiable if there are Lipschitz maps $f_i: \mathbf{R}^m \rightarrow \mathbf{R}^n$, $i = 1, 2, \dots$, such that

$$\mu \left(E \setminus \bigcup_{i=1}^{\infty} f_i(\mathbf{R}^m) \right) = 0.$$

A set A is called purely (μ, m) unrectifiable if $\mu(A \cap E) = 0$ for every (μ, m) rectifiable set E .

Inspecting the proof of Theorem 18.1 we find that it gives the following result.

Suppose μ is a Borel regular measure on $\mathbf{R}^n$ and A is a Borel set with $\mu(A) < \infty$ such that the following two conditions are satisfied for $\gamma_{n,m}$ almost all $V \in G(n, m)$:

- (i) $\text{card}(A \cap P_V^{-1}\{y\}) < \infty$ for $\mathcal{H}^m$ almost all $y \in V$.
- (ii) $\mathcal{H}^m(P_V B) = 0$ whenever $B \subset A$ and $\mu(B) = 0$.

If A is purely (μ, m) unrectifiable, then $\mathcal{H}^m(P_V A) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.

These conditions are satisfied for the integralgeometric measure $\mathcal{I}_\infty^m$ of 5.14, see Federer [3, 3.3.14]. Although $\mathcal{I}_\infty^m(B) = 0$ implies $\mathcal{H}^m(P_V B) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$, there is a problem with condition (ii): the exceptional set of V 's has to be independent of B . Since $\mathcal{I}_\infty^m$ is Borel regular and for Borel sets B , $\mathcal{I}_\infty^m(B) = 0$ if and only if $\mathcal{H}^m(P_V B) = 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$, the result for $\mathcal{I}_\infty^m$ can be restated in the form

$$\mathcal{I}_\infty^m(A) < \infty \text{ implies } A \text{ is } (\mathcal{I}_\infty^m, m) \text{ rectifiable.}$$

The corresponding result for $\mathcal{I}_1^m$ is false, see Mattila [10]. For $\mathcal{I}_t^m$, $1 < t < \infty$, its validity is unknown.

(3) In Theorem 18.1 we gave a criterion for rectifiability in terms of projections. In a way a more precise characterization can be given in terms of the integralgeometric measures:

If $1 \leq t \leq \infty$ and $\mathcal{H}^m(A) < \infty$, then A is m -rectifiable if and only if $\mathcal{H}^m(A) = c\mathcal{I}_t^m(A)$, where c is a suitable normalization constant depending only on m , n and t .

See Federer [3, 3.3.13]. In his book Federer also uses Carathéodory's construction to define many other natural m -dimensional measures and he shows that all of them agree for m -rectifiable sets.

(4) The sufficient condition in 18.1(2) for a set to be purely m -unrectifiable can be considerably strengthened. For example if Γ is a rectifiable curve in the plane $\mathbf{R}^2$ it is not too hard to show that any $\mathcal{H}^1$ measurable subset of Γ with positive $\mathcal{H}^1$ measure can project into a set of length zero in at most one direction. Hence if $A \subset \mathbf{R}^2$ is $\mathcal{H}^1$ measurable with $\mathcal{H}^1(A) < \infty$ and if we can find two lines $L_i \in G(2, 1)$, $i = 1, 2$, such that $\mathcal{H}^1(P_{L_i} A) = 0$ for $i = 1, 2$, we can conclude that A is purely 1-unrectifiable. Then from this information on only two projections Theorem 18.1 tells us much more: $\mathcal{H}^1(P_L A) = 0$ for $\gamma_{2,1}$ almost all lines $L \in G(2, 1)$. A corresponding result in general dimensions can be obtained from Federer [3, 3.2.27].

Note however that to conclude from finitely many projections of measure zero that almost all projections have measure zero we need to know that A has finite $\mathcal{H}^m$ measure. For example, if $C \subset \mathbf{R}$ is a Borel set with $\mathcal{L}^1(C) = 0$ and $\dim C > m/n$, then for $A = C \times \cdots \times C \subset \mathbf{R}^n$, the projection of A on every m -dimensional coordinate plane has $\mathcal{H}^m$ measure zero but as $\dim A > m$ by Theorem 8.10 $\mathcal{H}^m(P_V A) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$ by Corollary 9.8.

For Borel sets A of Hausdorff dimension m and of non- σ -finite $\mathcal{H}^m$ measure much stranger things can happen. Falconer [8] proved, see also Falconer [4, §7.3], [10] and [16], that it is possible essentially to give the projections on m -planes in advance and then find A with these projections, up to $\mathcal{H}^m$ null-sets. For $m = 1$ and $n = 2$ this follows with a dualization of a result of Davies [2]. Falconer uses a “Venetian blind” construction similar to that in 5.14 (Figure 5.2). Talagrand [1] has a related result for projections in $\mathbf{R}^2$.

(5) Let S be the one-dimensional Sierpinski gasket as in Figure 5.1. In Guzmán [2, p. 214] one finds a method due to Kahane to prove that $\mathcal{H}^1(P_L S) = 0$ for $\gamma_{2,1}$ almost all $L \in G(2, 1)$. However, it seems to be difficult to decide for which lines L this holds. Kenyon [2] showed that $\mathcal{H}^1(P_L S) = 0$ if the angle between L and the x -axis is irrational.

Applying Corollary 9.4 and Theorem 18.1 to self-similar sets such as S one obtains self-similar subsets K of $\mathbf{R}$ with $\dim K = 1$ and $\mathcal{L}^1(K) = 0$; see Exercise 2. It is not known whether there exist self-similar subsets K of $\mathbf{R}$ with $\dim K = s$ and $\mathcal{H}^s(K) = 0$ for other values of s , $0 < s < 1$, see Peres [2], recall also Exercises 9.3–5.

(6) Theorem 18.1 is interesting in itself but it has also played a very important role in the development of the theory of currents and geometric calculus of variations. When founding the theory of integral currents Federer and Fleming [1] proved the central compactness theorem using Theorem 18.1. Later other proofs have been given by Solomon [1], Almgren [2] and White [1]. Other applications of Theorem 18.1 have been given by Almgren [1] and Zheng [1].

Besicovitch sets

Theorem 18.1 can also be applied to curve packing problems. The most classical is the existence of the so-called Besicovitch sets. These are Borel subsets of $\mathbf{R}^2$ of Lebesgue measure zero but still containing a line in every direction. Besicovitch constructed such a surprising set in Besicovitch [2] (see Falconer [4], Guzmán [1]–[2] and Kahane [1] for this

and related constructions) but much later in Besicovitch [8] he observed that Theorem 18.1 gives them quite easily. We now use Theorem 18.1.

18.11. Theorem. *There is a Borel set $B \subset \mathbf{R}^2$ with $\mathcal{L}^2(B) = 0$ containing a line in every direction.*

We use Theorem 18.1 to find the following set, which will be used as the parameter set for our lines.

18.12. Lemma. *There is a compact set $C \subset \mathbf{R}^2$ such that $0 < \mathcal{H}^1(C) < \infty$, $\pi C = [0, 1]$, where $\pi(x, y) = x$, and $\mathcal{H}^1(P_L C) = 0$ for $\gamma_{2,1}$ almost all $L \in G(2, 1)$.*

Proof. All we need is a compact purely 1-unrectifiable set whose projection on the x -axis is $[0, 1]$. For example, we can take as C the self-similar Cantor set the first step of whose construction is in Figure 18.1. There are many ways to see that C is purely 1-unrectifiable. For instance, one easily verifies that it has no approximate tangents, or one finds two directions where it projects into a set of measure zero. $\square$

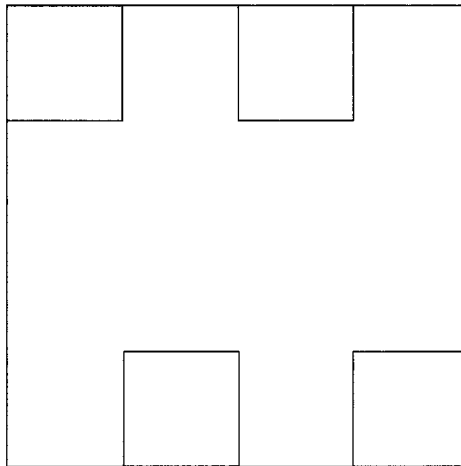


Figure 18.1.

de Guzmán observed that the method of Kahane presented in Guzmán [2, p. 214] can also be used to show that $\mathcal{H}^1(P_L A) = 0$ for $\gamma_{2,1}$ almost all $L \in G(2, 1)$ without referring to Theorem 18.1. In fact, for the proof of Theorem 18.11 the condition $\mathcal{H}^1(C) < \infty$ is not needed and thus also

for example the “Venetian blind” construction described in 5.14 and Falconer [16, 6.3] could be used to produce the desired set C .

We now prove Theorem 18.11. Using C as in Lemma 18.12 we consider the lines

$$\ell(a, b) = \{(x, y) : y = ax + b\}, \quad (a, b) \in C,$$

and define

$$B = \bigcup_{(a,b) \in C} \ell(a, b).$$

As $B = f(C \times \mathbf{R})$ where $f(a, b, x) = (x, ax + b)$, B is σ -compact, and in particular a Borel set. Since $\pi C = [0, 1]$ there is for every $a \in [0, 1]$ some $b \in \mathbf{R}$ such that $(a, b) \in C$, whence $\ell(a, b) \subset B$. Thus B contains a parallel of every line $y = ax$, $0 \leq a \leq 1$. Taking the union of four suitably rotated copies of B we get a Borel set containing a line in every direction. So all that is left is to show that $\mathcal{L}^2(B) = 0$.

By Fubini's theorem it is enough to show that almost every vertical line meets B in a set of $\mathcal{H}^1$ measure zero. For any $t \in \mathbf{R}$

$$\begin{aligned} B \cap \{(x, y) : x = t\} &= \bigcup_{(a,b) \in C} \ell(a, b) \cap \{(t, y) : y \in \mathbf{R}\} \\ &= \{(t, at + b) : (a, b) \in C\} = \{t\} \times \pi_t(C), \end{aligned}$$

where $\pi_t(x, y) = tx + y$. The map π_t is almost a projection. That is, if $\theta = (\theta_1, \theta_2) \in S^1$ with $\theta_2 \neq 0$ and L_θ is the line through 0 and θ , then $P_{L_\theta}(x, y) = \pi_{\theta_1/\theta_2}(x, y)\theta_2\theta$. It follows that $\mathcal{L}^1(\pi_t(C)) = 0$ for $\mathcal{L}^1$ almost all $t \in \mathbf{R}$ if and only if $\mathcal{H}^1(P_{L_\theta}C) = 0$ for $\mathcal{H}^1$ almost all $\theta \in S^1$. Since we have the latter, $\mathcal{H}^1(B \cap \{(x, y) : x = t\}) = 0$ for $\mathcal{L}^1$ almost all $t \in \mathbf{R}$ as required. $\square$

18.13. Remarks. (1) It is possible to use similar arguments for other curve packing problems, too. For example, one can find circles in the plane centred at every point of the interval $\{(a, 0) : 0 \leq a \leq 1\}$ such that their union has zero area; see Talagrand [1]. To do this, let

$$\begin{aligned} S(a, b) &= \{(x, y) : (x - a)^2 + y^2 = a^2 + b\} \\ &= \{(x, y) : y^2 = 2ax + b - x^2\}. \end{aligned}$$

Letting, with C as above,

$$B = \bigcup_{(a,b) \in C} S(a, b),$$

we have

$$B \cap \{(x, y) : x = t\} = \{t\} \times \{(2ta + b - t^2)^{1/2} : (a, b) \in C\},$$

which has $\mathcal{H}^1$ measure zero if and only if

$$\mathcal{L}^1(\{2ta + b : (a, b) \in C\}) = 0.$$

Hence $\mathcal{L}^2(B) = 0$ follows as above.

(2) There is also a Borel set of measure zero in $\mathbf{R}^2$ containing a circle of every radius; see Exercise 3. However, if $A \subset \mathbf{R}^2$ is Lebesgue measurable with $\mathcal{L}^2(A) > 0$ and if r is a positive function on A , then the union of the circles $S(x, r(x))$, $x \in A$, has positive Lebesgue measure. This follows from the work of Bourgain [1] and Marstrand [6]. An analogous result for spheres in $\mathbf{R}^n$, $n \geq 3$, is also true and easier, see Stein and Wainger [1]. The papers by Bourgain and by Stein and Wainger contain much more information about the spherical maximal function, $\sup_{r>0} r^{1-n} \int_{S(x,r)} f d\mathcal{H}^{n-1}$, yielding the following differentiation theorem over spheres:

If $f: \mathbf{R}^n \rightarrow \mathbf{R}$ is locally in L^p for some $p > n/(n-1)$, then

$$\lim_{r \downarrow 0} \mathcal{H}^{n-1}(S^{n-1})^{-1} r^{1-n} \int_{S(x,r)} f d\mathcal{H}^{n-1} = f(x)$$

for $\mathcal{L}^n$ almost all $x \in \mathbf{R}^n$. This is false for $p \leq n/(n-1)$. In particular, for $n = 2$ this is false for $p = 2$ which makes the problem more difficult in $\mathbf{R}^2$ than in higher dimensions.

For other results on curve packing problems, see e.g. Falconer [4, § 7] and Sawyer [1].

(3) The problem about the existence of higher dimensional Besicovitch sets is partially open. We say that a Borel set B is a Besicovitch (n, m) set if $B \subset \mathbf{R}^n$, $\mathcal{L}^n(B) = 0$ and B contains a translate of every $V \in G(n, m)$. Rotating Besicovitch $(2, 1)$ sets in $\mathbf{R}^n$ one can construct Besicovitch $(n, 1)$ sets for any n , $n \geq 2$. Marstrand [5] showed that there are no Besicovitch $(3, 2)$ sets, Falconer [1] proved the same about Besicovitch (n, m) sets for $m > n/2$ and Bourgain [3] for $2^{m-1} + m \geq n$. Falconer's attempt in [2] to show that no Besicovitch (n, m) sets exist for $m \geq 2$ contains an error and the problem is still open.

The Besicovitch sets in $\mathbf{R}^2$ cannot be very small; Davies [4] showed that their Hausdorff dimension is always 2, see also Falconer [4, Theorem 7.9]. His proof does not work in $\mathbf{R}^n$ for $n \geq 3$ and there the problem is open. Bourgain [3] has obtained some partial results. For example, he showed that if B is a Besicovitch $(3, 1)$ set, then $\dim B \geq 7/3$. This question, as Bourgain's work reveals, seems to be very closely connected with the behaviour of Fourier transforms on spheres.

Exercises.

1. Let A be a subset of some rectifiable curve in $\mathbf{R}^2$. Show that if $\mathcal{H}^1(A) > 0$ there can be at most one line $L \in G(2, 1)$ such that $\mathcal{H}^1(P_L A) = 0$.
2. Use Corollary 9.4 and Theorem 18.1 to find self-similar sets $K \subset \mathbf{R}$ with $\dim K = 1$ and $\mathcal{L}^1(K) = 0$. Hint: Use for example the set $E = C(1/4) \times C(1/4)$, recall 4.10, and show that the projections of E are also self-similar.
3. Construct a Borel set $B \subset \mathbf{R}^2$ such that $\mathcal{L}^2(B) = 0$ and that for every $r > 0$ there is $x \in \mathbf{R}^2$ such that $S(x, r) \subset B$; see Falconer [4, Theorem 7.10].
4. Show that there exists a Borel set $B \subset \mathbf{R}^2$ such that $\mathcal{L}^2(\mathbf{R}^2 \setminus B) = 0$ and for every $L \in G(2, 1)$ the projection $P_L(B)$ has no interior points in L . Hint: Use Besicovitch sets.
5. Show that there exist Besicovitch $(n, 1)$ sets for $n \geq 2$.